

Mixture continuity $\wedge$ Completeness $\Rightarrow$ Archimedean

Result — ai; produced by Claude Fable 5.1 (Cowork session).

0.1 Premises

- **Mixture continuity.** For all $p, q, r \in \Delta(X)$ the sets $\{\lambda \in [0,1] : \lambda p + (1-\lambda)q \geq r\}$ and $\{\lambda \in [0,1] : r \geq \lambda p + (1-\lambda)q\}$ are closed.
- **Completeness.** For all $p, q \in \Delta(X)$: $p \geq q$ or $q \geq p$.

0.2 Conclusion

- **Archimedean.** For all $p, q, r \in \Delta(X)$: if $p > q > r$ then there are $\alpha, \beta \in (0,1)$ with $\alpha p + (1-\alpha)r > q$ and $q > \beta p + (1-\beta)r$.

0.3 Proof

Let $p > q > r$ and set $A = \{\lambda : \lambda p + (1-\lambda)r \geq q\}$, $B = \{\lambda : q \geq \lambda p + (1-\lambda)r\}$. Both are closed by continuity and $A \cup B = [0,1]$ by completeness. Since $1 \notin B$ (as $p > q$) and $[0,1] \setminus B$ is open, some interval $(1-\varepsilon, 1]$ lies in $[0,1] \setminus B \subseteq A$; any α in it with $\alpha < 1$ gives $\alpha p + (1-\alpha)r > q$. Symmetrically, $0 \notin A$ (as $q > r$) yields $[0, \varepsilon) \subseteq B \setminus A$ and any β in it with $\beta > 0$ gives $q > \beta p + (1-\beta)r$.

0.4 Notes

Completeness is used to conclude $\lambda \in A$ from $\lambda \notin B$; whether continuity alone suffices is left open here.

Record: topics/decision-theory/results/continuity-completeness-imply-archimedean.yaml